

**THE UNITED REPUBLIC OF TANZANIA
NATIONAL EXAMINATIONS COUNCIL OF TANZANIA
ADVANCED CERTIFICATE OF SECONDARY EDUCATION
EXAMINATION**

142/2

**ADVANCED MATHEMATICS 2
(For Both School and Private Candidates)**

Duration: 3 Hours

Year: 2026

Instructions

1. This paper consists of sections A and B with a total of **eight (8)** questions.
2. Answer **all** questions in section A and **two (2)** questions from section B.
3. All work done in answering each question must be shown clearly.
4. NECTA's Mathematical tables and non-programmable calculators may be used.
5. All writing must be in **black** or **blue** ink, **except** drawings which must be in pencil.
6. You should **not** write any notes in this question paper.
7. Communication devices and any unauthorised materials are **not** allowed in the examination room.
8. Write your **Examination Number** on every page of your answer booklet(s).

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SECTION A (60 Marks)

Answer all questions in this section.

1. (a) (i) A man has three sorts of shirts with red, blue and green colours and two pairs of trousers coloured black and white. How many different outfits can the man match if each outfit consists of one shirt and one pair of trouser?
- (ii) The following table shows the probability distribution of a random variable X . Find the expectation of X and standard deviation of X .

x	1	2	3	4	5
$P(X = x)$	0.1	0.3	0.3	0.2	0.1

- (b) A box contains 500 bolts. The probability that there is defective bolt is 0.006. Find the probability that the box contains 3 defective bolts.
- (c) Find the variance of a continuous random variable X , which has a probability

$$\text{density function, } f(x) = \begin{cases} \frac{6}{7}x & \text{for } 0 \leq x \leq 1 \\ \frac{6}{7}(2x - x^2) & \text{for } 1 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

2. (a) Use the laws of algebra of propositions to simplify the argument $[\sim p \wedge (q \vee \sim p)] \vee [\sim p \rightarrow \sim q]$.
- (b) Draw circuit diagram representing the expressions $(P \wedge R) \vee (\sim P \wedge \sim Q)$.
- (c) Use truth table to test the validity of the following argument:
 "If I go to the market then I cannot go to school. Either I cannot go to school or I pass my sports competitions. If I pass my sports competitions then I can go to school. Therefore, if I go to school then I cannot go to the market."
3. (a) The vectors u and v are such that $|u| = 5$ and $|v| = 8$ and $u \cdot v = 20$. Find the value of the constant p such that $u + pv$ is perpendicular to u .
- (b) The constant force $\underline{F} = 10\underline{i} + 18\underline{j} - 6\underline{k}$ moves an object along a straight line from point $P(2, 3, 0)$ to $Q(4, 9, 15)$. If the distance from P to Q is measured in

metres and the magnitude of the given force is in Newtons, find the work done in Joules.

- (c) The acceleration of a particle at point P is given by the vector $(\underline{i} + \underline{j}) \text{ m/s}^2$. At time $t=0$ the particle passes through the origin with velocity $(\underline{i} - \underline{j}) \text{ m/s}$. Determine the position vector and velocity of the particle at $t=2$.

4. (a) The complex number $z_1 = \frac{c}{1+i}$ and $z_2 = \frac{d}{1+2i}$ where $c, d \in \mathbb{R}$ and such that $z_1 + z_2 = 1$, find the values of c and d .

- (b) (i) Given $z = x + iy$, find the Cartesian equation if $\arg(z+i) = \frac{\pi}{4}$.

- (ii) Express $\sqrt{1+i}$ in polar form.

- (c) Solve $z^2 - 1 - \sqrt{3}i = 0$ giving the values of z in the form of $re^{i\theta}$ where $r \in \mathbb{R}$ and $-\pi \leq \theta \leq \pi$.

SECTION B (40 Marks)

Answer two (2) questions from this section.

5. (a) If A , B and C are angles of a triangle, prove that $\sin A + \sin B + \sin C = 4 \cos\left(\frac{A}{2}\right) \cos\left(\frac{B}{2}\right) \cos\left(\frac{C}{2}\right)$.

- (b) Solve the equation $\cos 6x + \cos 4x + \cos 2x = 0$ for $0^\circ \leq x \leq 180^\circ$.

- (c) (i) Sketch the graph of $f(\theta) = 2 \sin \theta$ in the interval $-\pi \leq \theta \leq \pi$.

- (ii) Compute the value of $\int_0^{\frac{\pi}{8}} \cos^2 x dx$.

6. (a) Solve for x in the equation $3^{2x-1} = 3^x + 18$.

- (b) (i) Use the method of induction to prove that $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$.

- (ii) A ball bounces to the displacement equals to $\frac{2}{3}$ of the displacement covered in the previous bounce. If the ball is thrown to a height of 25 feet, determine the distance it would have moved when it stops.

(c) (i) Given that $A = \begin{pmatrix} 2 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 1 & 2 \\ 3 & 2 & 2 \\ 1 & 0 & 1 \end{pmatrix}$, compute AB .

- (ii) Use the property of the matrices observed in part (c) (i) to solve the following simultaneous equations:

$$\begin{cases} 2x - y - 2z = 6 \\ -x + y = 1 \\ -2x + y + 3z = -7 \end{cases}$$

7. (a) If $y^2 = 4bx$, formulate the first order differential equation.
- (b) Find the general solution of the differential equation $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 6e^x + \sin x$.
- (c) A tank contains a homogeneous solution of salt and water. Initially the tank contains 500 litres of water with 1.0 kg of salt dissolved in it. The mixture is poured off at a rate of 10 litres per minute. Find the kilograms of salt in the tank after 2 minutes.
8. (a) Find the vertex, focus and length of the latus rectum of the parabola $y^2 - 2y - 4x - 19 = 0$.
- (b) Verify that $bx \sec \theta - ay \tan \theta = ab$ is the equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at $P(a \sec \theta, b \tan \theta)$.
- (c) An ellipse has its foci on the y-axis and its centre at the origin. If the distance between the foci is 6 units and the length of the major axis is 10 units, determine the eccentricity, the equations of directrices and the equation of the ellipse.
- (d) (i) Express $r = d \sec(\theta - \alpha)$ in Cartesian form.
- (ii) Find the polar equation of the Cartesian equation $(x-a)^2 + (y-a)^2 = a^2$.